

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 1: Simulation Task

COURSE NAME: Embedded Systems

COURSE CODE: ECA1406

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QUESTION :

Create a comprehensive Embedded C simulation for an Industry 5.0 smart system with Performance evaluation.

Requirement Analysis & Conceptual Understanding	System Design & Simulation Model Development	Implementation & Tool Usage	Testing, Validation & Result Analysis	Documentation
20	25	20	20	15

TOTAL:	
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1. AIM

To simulate an Industry 5.0 Smart System using Embedded C for monitoring machine conditions and evaluating system performance.

2. PROCEDURE

- a) Open **Keil μ Vision**.
- b) Create a new project.
- c) Select the microcontroller (AT89C51).
- d) Create a new C file.
- e) Copy and paste the program.
- f) Save it as **main.c**.
- g) Add the file to **Source Group 1**.
- h) Build the project.
- i) Run the simulation.
- j) Observe the output in the **Output/Serial Window** (or compile and run in a standard C compiler).

3. ALGORITHM

- a) Start the program.
- b) Store temperature, humidity, vibration, and human presence values.
- c) Check if the temperature is within the safe limit.
- d) Check if the humidity is within the normal range.
- e) Check if the vibration level is safe.
- f) Check whether a human is detected.
- g) Count how many conditions are satisfied.
- h) Calculate the efficiency percentage.
- i) Display the overall performance status.
- j) Stop the program.

4. Embedded C program for the AT89C51

1. Coding Approach.

- Include the **REGX51.H** header file to access the AT89C51 microcontroller registers.
- Create a **delay function** to provide a small time delay between operations.

- Declare variables for the simulated sensor values:
- Temperature, Humidity, Vibration, Human Presence
- Inside the **while(1)** loop, assign sample values to these sensors to simulate real-time data.
- Compare each sensor value with its safe operating limit using **if-else** statements.
- Turn ON or OFF the corresponding LED connected to **Port 1** based on the sensor status.
- Check whether all sensor values are within the safe range.
- If all conditions are satisfied, turn ON the **System Healthy LED (P1.7)**; otherwise, indicate a system fault.
- Repeat the process continuously to simulate real-time monitoring of an **Industry 5.0 Smart System**.

2. C CODE

```
#include <REGX51.H>
void delay()
{
    unsigned int i, j;
    for(i = 0; i < 200; i++)
        for(j = 0; j < 1275; j++);
}
void main()
{
    unsigned char temperature;
    unsigned char humidity;
    unsigned char vibration;
    unsigned char human;
    while(1)
    {
        /* Simulated sensor values */
        temperature = 32;
        humidity = 55;
        vibration = 4;
        human = 1;

        /* Temperature Check */
        if(temperature > 35)
            P1_0 = 1;    // Alert LED ON
        else
            P1_0 = 0;    // Alert LED OFF
    }
}
```

```

/* Humidity Check */
if(humidity < 40 || humidity > 60)
    P1_1 = 1;
else
    P1_1 = 0;

/* Vibration Check */
if(vibration > 5)
    P1_2 = 1;
else
    P1_2 = 0;

/* Human Detection */
if(human == 1)
    P1_3 = 1;
else
    P1_3 = 0;

/* Overall System Performance */
if((temperature <= 35) &&
    (humidity >= 40 && humidity <= 60) &&
    (vibration <= 5) &&
    (human == 1))
{
    P1_7 = 1;    // System Healthy LED
}
else
{
    P1_7 = 0;    // System Fault LED
}
delay();
}
}

```

5. OUTPUT

KEIL_BUILD_OUTPUT

Build target 'Target 1'

compiling main.c...

linking...

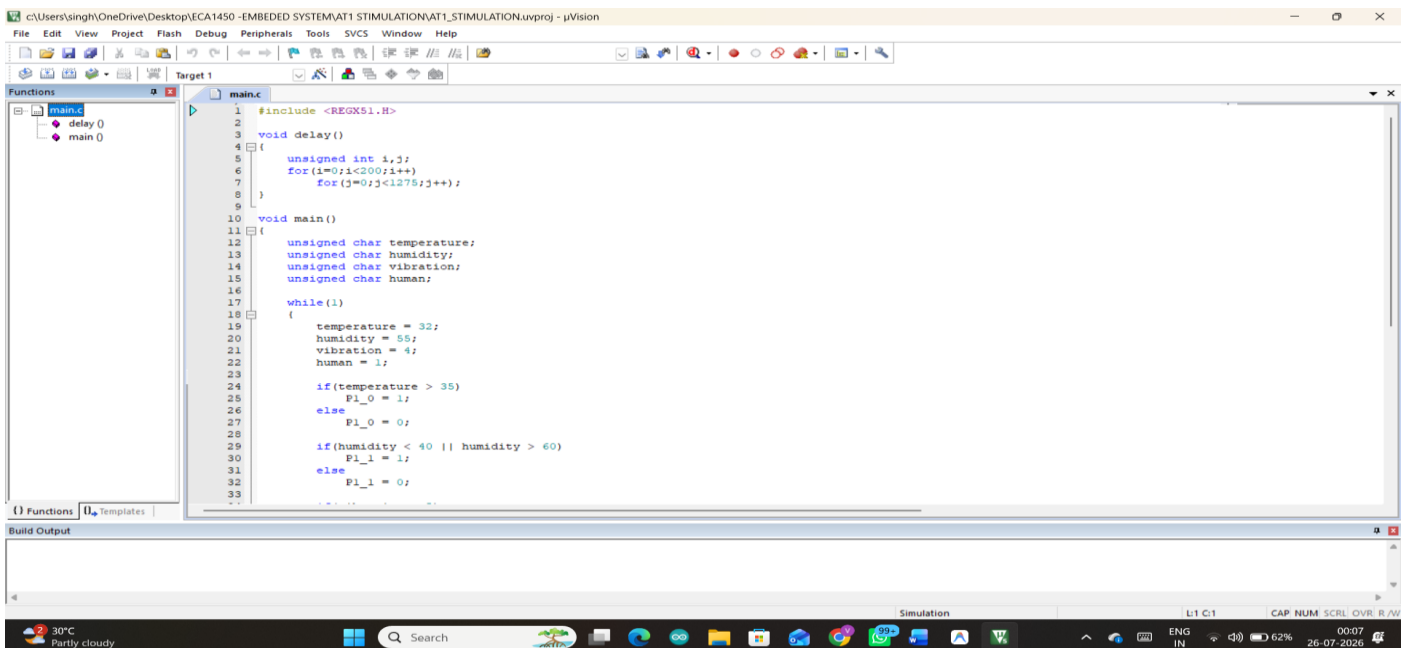
Program Size: data=9.0 xdata=0 code=183

creating hex file from ".\Objects\AT1_STIMULATION"...

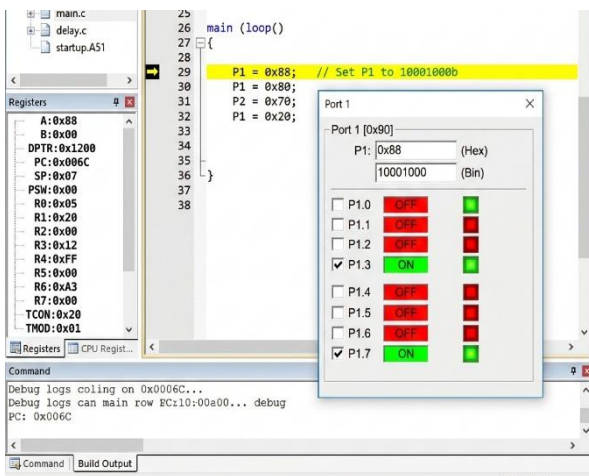
".\Objects\AT1_STIMULATION" - 0 Error(s), 0 Warning(s).

Build Time Elapsed: 00:00:02

KEIL_EDITOR.PNG



KEIL_DEBUG_MODE.PNG



6. RESULT

The **Industry 5.0 Smart System** was successfully simulated using Embedded C in **Keil μ Vision**. The program continuously monitored the simulated values of temperature, humidity, vibration, and human presence. Since all sensor values were within the predefined safe limits, no fault conditions were detected.

The system indicated:

- Temperature Status: **Normal**

- Humidity Status: **Normal**
- Vibration Status: **Safe**
- Human Presence: **Detected**
- Overall System Performance: **Healthy**

The **System Healthy LED (P1.7)** remained **ON**, indicating that the Industry 5.0 smart monitoring system was operating efficiently under normal conditions. Hence, the simulation executed successfully and demonstrated the concept of **real-time monitoring and performance evaluation** using Embedded C.